

Incidence toric ideals and three-point functions

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Joint work with Barbara Betti, Sean Grate and Flavio Salizzoni

A physics motivation

- ① n massless particles $p_1, \dots, p_n$ in $\mathbb{R}^d$
- ② $p_{ij} = p_i \cdot p_j$ (Lorentzian inner product).
- ③ $P = (p_{ij})$ is a symmetric $n \times n$ matrix with zero diagonal.
- ④ Simplified form of 3-point functions: $C_{ijk} = \frac{1}{p_{ij}p_{ik}p_{jk}}$ (topology?)
- ⑤ Relations between C_{ijk} are the same as relations between $c_{ijk} = p_{ij}p_{ik}p_{jk}$ (toric ideals!)

$$\varphi_{n,3,2} : \mathbb{C}[c_{ijk} : 1 \leq i < j < k \leq n] \longrightarrow \mathbb{C}[p_{ij} : 1 \leq i < j \leq n]$$

$$\varphi_{n,3,2}(c_{ijk}) = p_{ij}p_{ik}p_{jk}$$

Incidence toric ideals

Let $n > k > t$ be integers. The incidence toric ideal $I_{n,k,t}$ is the kernel of the map

$$\varphi_{n,k,t} : \mathbb{C}[c_A : A \in \binom{[n]}{k}] \longrightarrow \mathbb{C}[p_B : B \in \binom{[n]}{t}]$$

$$\varphi_{n,k,t}(c_A) = \prod_{B \in \binom{A}{t}} p_B$$

The matrix associated to $\varphi_{n,k,t}$ is the incidence matrix of k and t -subsets of $[n]$

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$I_{n,2,1}$ is the toric edge ideal of the complete graph K_n

What is a bipartite hypergraph?

Theorem (Villarreal 95, Ohsugi-Hibi 99)

*The ideal $I_{n,2,1}$ is generated by binomials given by **primitive even closed walks** in K_n .*

Theorem (Petrović and Stasi, 2014)

*If H is a hypergraph, a Markov basis of the toric ideal I_H is given by **monomial walks** in H .*

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Toric ideals of graphs map edges to vertices. This is analogous to $I_{n,k,k-1}$

Question

What is the right notion of "even" for incidence toric ideals?

Answer?: **balancedness in topology!**

Boundary maps without signs

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

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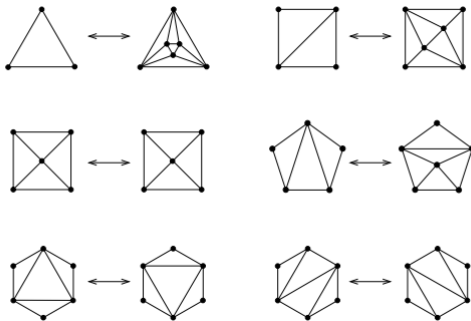
Theorem (Betti-Grate-H-Salizzoni, 2026+)

Let Δ be a triangulation of an orientable d -manifold such that *the graph of Δ is $(d+1)$ -colorable*. Then the orientation of Δ gives a primitive binomial in $I_{n,d+1,d}$.

Balanced bistellar flips: how to find elements in $I_{n,k,t}$

Question

If we know all the elements in $I_{n_0,k,t}$, how can we generate **new** elements in $I_{n,k,t}$ for $n > n_0$?



(Izmestiev, Klee, Novik 2018)

An intuition for elements in $I_{n,k,t}$ that are not manifolds



We can glue two nonbalanced manifolds and get a binomial in $I_{n,k,t}$, just as we can glue two odd cycles and get an element in $I_{n,2,1}$

The dimension: same matrix, different interpretations

- 1 Design theory (**null-designs**)
- 2 Algebraic geometry (**Hard Lefschetz theorem** for $\mathbb{P}^1 \times \dots \times \mathbb{P}^1$)
- 3 Symmetric polynomials (**Specht polynomials** of shape $(n - k, k)$)
- 4 and more

Theorem (Betti-Grate-H-Salizzoni, 2026+)

For every $n > k > t$, then

$$\dim \frac{R}{I_{n,k,t}} = \min \left(\binom{n}{k}, \binom{n}{t} \right)$$

Proof idea: the corresponding matrix always has full rank

A polytope and some of its combinatorics

Let $\mathcal{P}_{n,k,t}$ be the convex hull of the columns of the incidence matrix of k and t -subsets of $[n]$.

$$\begin{array}{c} 123 \quad 124 \quad 134 \quad 234 \\ \begin{array}{c} 12 \\ 13 \\ 23 \\ 14 \\ 24 \\ 34 \end{array} \left[\begin{array}{cccc} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right] \end{array}$$

$$n = 4, k = 3, t = 2$$

Theorem (Betti-Grate-H-Salizzoni, 2026+)

The polytope $\mathcal{P}_{n,k,t}$ is $(2^t - 1)$ -neighborly. In other words, every set of $2^t - 1$ vertices of $\mathcal{P}_{n,k,t}$ forms a simplicial face of $\mathcal{P}_{n,k,t}$.

Proof idea: Identify kernel elements with null-designs

A polytope and some of its geometry

Theorem (Betti-Grate-H-Salizzoni, 2026+)

Let f be the function such that $f(n, k, t) = \text{Vol}(\mathcal{P}_{n,k,t})$, where Vol is the normalized volume with respect to the euclidean lattice. Then f is not a polynomial function.

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$$\text{Vol}(\mathcal{P}_{6,3,2}) = 162$$

$$\text{Vol}(\mathcal{P}_{7,3,2}) = 85368$$

but our methods only give $6 \leq \text{Vol}(\mathcal{P}_{7,3,2})$

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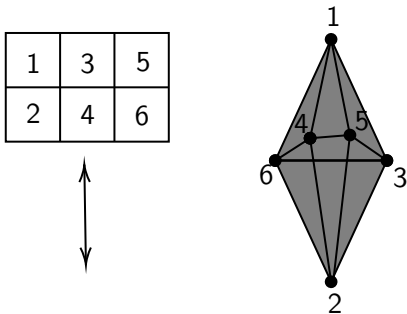
Let $J_{n,k}$ be the ideal generated by the binomials arising from every $(k-1)$ -dimensional crosspolytope on vertex set $A \subseteq [n]$. Then

$$J_{n,k} = I_{n,k,k-1} : m^\infty = \{f : fm^b \in I_{n,k,k-1} \text{ for some } b\},$$

where m is the product of all variables $c_{ij\ell}$.

Proof idea: orientations of crosspolytopes are known to generate kernel of incidence matrix between k and $k-1$ subsets of $[n]$

A combinatorial way of viewing crosspolytopes and $J_{2k,k}$: Specht polynomials of (k, k)



$$(x_1 - x_2)(x_3 - x_4)(x_5 - x_6)$$

$$x_1 x_3 x_5 + x_2 x_4 x_5 + x_2 x_3 x_6 + x_1 x_4 x_6 - x_2 x_3 x_5 - x_1 x_4 x_5 - x_1 x_3 x_6 - x_2 x_4 x_6$$

$$c_{135} c_{245} c_{236} c_{146} - c_{235} c_{145} c_{136} c_{246} \in J_{6,3}$$

Check out the preprint!

